A

Mini Project Report

On

ALCOHOL DETECTION ENGINE LOCK

Submitted for partial fulfilment of the requirements for the award of the degree of

BACHELOR OF ENGINEERING

In

ELECTRONICS AND COMMUNICATION

By

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. Vyshnavi (2451-22-735-060) in partial fulfilment of the requirements for the award of degree of Bachelor of Engineering in Electronics and Communication from Maturi Venkata Subba Rao (M.V.S.R) Engineering College, an Autonomous Institution, affiliated to Osmania University Hyderabad, during the Academic Year 2024 – 2025 under the guidance and supervision

The results embodied in this report have not been submitted to any other university or institute for the award of any degree or diploma.

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This is to certify that the work reported in the present mini project entitled "<project title>" is a record of Bonafide work done by us in the department of Electronics and Communication Engineering, Maturi Venkata Subba Rao (M.V.S.R.) Engineering College, an Autonomous Institution, affiliated to Osmania University. The reports are based on project work done by us entirely by us and not copied from any other source. The results embodied in this project report have not been submitted to any University or Institute for the award of any degree or diploma to the best of our knowledge and belief.

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ABSTRACT

Drunk driving is a major contributor to road accidents worldwide. This project proposes a cost-effective and efficient solution to prevent drunk driving using an Arduino-based smart engine lock system.

The system integrates an alcohol sensor, GPS module, and GSM module to detect and prevent drunk driving. The alcohol sensor detects the driver's blood alcohol concentration (BAC) and sends the data to the Arduino microcontroller. If the BAC exceeds the predetermined threshold, the engine is automatically locked, preventing the vehicle from starting. The GPS module tracks the vehicle's location and sends coordinates to a central server or authorized personnel via the GSM module.

The system sends SMS alerts to pre-programmed numbers, notifying authorities or family members of potential drunk driving incidents. The proposed system promotes road safety and protects lives.

The system's performance is evaluated through experiments, demonstrating its accuracy and effectiveness. The proposed system has the potential to be integrated into modern vehicles, providing a safer driving experience.

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CHAPTER 1

1.INTRODUCTION

The concept described in this project report is aimed to regret the drunken driver by not allowing driving the vehicle. When this system is installed in a vehicle over the dashboard and detects the drunken driver through the alcohol sensor, the simulated vehicle presented here with a DC motor, which will be stopped automatically, treated as engine shutdown. Since the availability of the exact sensor is not possible and since it is a prototype module, the basic concept is proven with the universal sensor which can detect all sorts of toxic gases, petroleum products, smoke, etc., in addition, to the alcoholic vapor.

The sensor used here is named MQ6. Since it can detect all sorts of toxic vapor and is available easily everywhere, it is said to be the universal sensor. This sensor is used here to detect alcoholic vapours. Presently, this kind of sensor can be used for goods transport vehicles, because these vehicles will not carry passengers. At the point when this sensor is utilized in vehicles, it is hard to recognize the tanked individual in light of the tipsy travellers. Henceforth it is prescribed to involve this innovation in trucks just because the truck contains a separate lodge for the driver and his associate. If a proper sensor is used, it can be installed in cars also. One benefit of utilizing this sensor is that it cannot distinguish minimal far fumes, since it isn't the case delicate. It is supposed to be an advantage since it shouldn't identify other tipsy people the individuals who are minimum away from the driving wheel.

This shows that the sensor should be introduced over the dashboard and that direction should be extremely close toward the driving wheel. Assuming that this sort of course of action is made in the lodge, the framework can identify just a tipsy driver. As portrayed over, the sensor isn't delicate, during a demo, the sensor should be presented to the liquor fume. For this purpose, pour little alcohol (brandy or whiskey) into a small cup and place the sensor a little above the cup with a gap of 2 to 3cms approximately. Contingent upon the liquor fume focus in the air, the conductivity of the sensor will be changed and in light of this conductivity, outplaced in the structure of voltage levels will be varied automatically.

1.1 PROBLEM STATEMENT

Driving under the influence of alcohol is a major cause of road accidents, and vehicle theft is a persistent issue. Traditional ignition systems can be bypassed, making vehicles vulnerable to unauthorized use. This project aims to develop an Arduino-based alcohol detection system that prevents a vehicle from starting if alcohol is detected in the driver's breath. Using an alcohol sensor (e.g., MQ-3) integrated with an Arduino microcontroller, the system will lock the engine if alcohol levels exceed a safe threshold, enhancing road safety and vehicle security by ensuring only sober individuals can operate the vehicle.

1.2 WORKING PRINCIPLE

The alcohol detector and engine locking system is a safety feature that is designed to prevent drivers from operating their vehicles if their blood alcohol content (BAC) level is above a certain limit. The system typically consists of a Breathalyzer device that measures the driver's BAC level and a computer system that controls the engine and ignition.

When a driver enters the vehicle, they are required to blow into the Breathalyzer device. The device measures the alcohol content in their breath and calculates their BAC level. If their BAC level is above the set limit (usually 0.08% in the United States), the engine will not start. If the drive attempts to start the engine multiple times and continues to have a high BAC level, the system may trigger an alarm or notify law enforcement.

1.3 SCOPE

The project aims to design and develop an intelligent engine lock system that detects the driver's blood alcohol concentration (BAC) using an alcohol sensor and prevents the vehicle from starting if the BAC exceeds a predetermined threshold. The system will utilize an Arduino microcontroller, GPS module, and GSM module to provide a comprehensive solution for drunk driving prevention.

CHAPTER 2

2.1 LITERATURE SURVEY

Here is a literature survey on alcohol detection with engine locking system using GPS and Wi-Fi module: Alcohol Detection System with Engine Locking using GPS and GSM" by G. G. Dahake and R. S. Bichkar. This paper discusses the design and implementation of an alcohol detection system with engine locking using GPS and GSM. The system uses a gas sensor to detect alcohol and sends an alert to the owner's phone through GSM if alcohol is detected.

"Development of a Smart Car Alcohol Detection System Based on GPS and GSM Technologies" by T. R. Liu et al. This paper describes the development of a smart car alcohol detection system based on GPS and GSM technologies. The system uses a gas sensor to detect alcohol and sends a message to the owner's phone through GSM if alcohol is detected.

"Design and Implementation of an Alcohol Detection System with Engine Locking Using GPS and Wi-Fi" by N. A. Bhat and V. D. Shinde. This paper describes the design and implementation of an alcohol detection system with engine locking using GPS and Wi-Fi. The system uses a gas sensor to detect alcohol and sends a signal to the engine control unit to prevent the engine from starting if alcohol is detected.

"Alcohol Detection System with Engine Locking Using GPS and Wi-Fi" by S. S. Patil et al. This paper discusses the design and implementation of an alcohol detection system with engine locking using GPS and WIFI. The system uses a gas sensor to detect alcohol and sends a signal to the engine control unit to prevent the engine from starting if alcohol is detected.

Real-Time Alcohol Detection System with Engine Locking Using GPS and Wi-Fi" by S. D. Kore and P. D. Dumbre. This paper presents the design and implementation of a real-time alcohol detection system with engine locking using GPS and Wi-Fi. The system uses a gas sensor to detect alcohol and sends a signal to the engine control unit to prevent the engine from starting if alcohol is detected.

In summary, these papers highlight the importance of alcohol detection systems with engine locking using GPS and Wi-Fi modules in preventing drunk driving and ensuring road safety. They discuss the design, implementation, and testing of such systems, demonstrating their effectiveness in detecting alcohol and preventing vehicle operation.

2.2 EXISTING SOLUTION

In this existing system, alcohol detectors are not proposed inbuilt in a car. The traffic police uses alcohol detectors (device) to avoid drink and drive accidents. The limitation of this device is that the police are not able to check each & every car & even if they stop some suspects there are chances that the police can be easily bribed. Therefore, there is a need for an effective system to check drunken drivers. In previous technologies can do only detecting the alcohol, If the person drinks the alcohol the engine will be locked.

2.3 PROPOSED SOLUTION

In Our proposed solution MQ3 alcohol sensor will continuously monitoring the alcohol. When the alcohol level reaches the trigger point then the MQ3 sensor sends the signal to the Arduino and the engine will be automatically turned off.

CHAPTER 3

3. SYSTEM REQUIRMENTS AND SPECIFICATIONS

3.1 HARDWARE SPECIFICATIONS

3.1.1 Arduino (UNO)

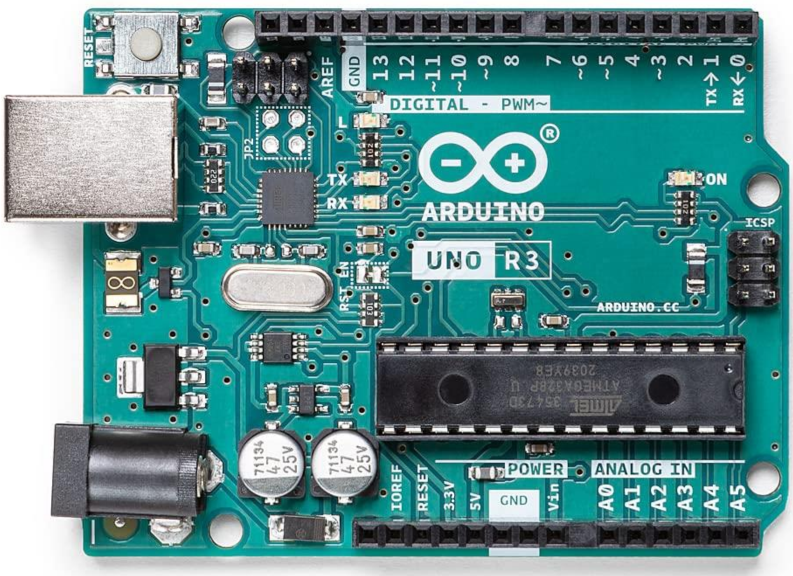

Fig 3.1.1: Arduino UNO

The Arduino Uno is an open-source microcontroller board based on the Microchip ATmega328P microcontroller and developed by Arduino.cc and initially released in 2010. The board as shown in fig 2.1.1 is equipped with sets of digital and analog input/output (I/O) pins that may be interfaced to various expansion boards (shields) and other circuits. The board has 14 digital I/O pins (six capable of PWM output), 6 analog I/O pins, and is programmable with the Arduino IDE (Integrated Development Environment), via a type B USB cable. It can be powered by the USB cable or by an external 9-volt battery, though it accepts voltages between 7 and 20 volts. It is similar to the Arduino Nano and Leonardo. The hardware reference design is distributed under a Creative Commons Attribution Share-Alike 2.5 license and is available on the Arduino website. Layout and production files for some versions of the hardware are also available. The hardware structure of Arduino Uno consists of:

1. Microcontroller

2. 14 Digital Pin

3. 6 Analog Pin

4. Power Supply

5. Power Jack

6.USB Port

7. Reset Button

Information about the hardware components:

• Microcontroller: Microcontroller is the central processing unit of Arduino Uno.

• Digital Pins: There are 14 digital pins on Arduino Uno which can be connected to components like LED, LCD, etc.

• Analog Pins: There are 6 analog pins on the Uno. These pins are generally used to connect sensors because all the sensors generally have analog values. Most of the input components are connected here.

• Power Supply: The power supply pins are IOREF, GND, 3.3V, 5V, Vin are used to connecting sensors because all the sensors generally have analog values. Most of the input components are connected here.

• Power Jack: Uno board can be powered both by external supply and via USB cable.

• USB Port: This port function is to program the board or to upload the program. The program can be uploaded to the board with the help of Arduino IDE and USB cable.

Reset Button: This is used to restart the uploaded program.

3.1.2 DC motor

Fig 3.1.2: DC motor

A DC Motor is an electrical device that converts electrical energy into mechanical energy. Going by the DC motor full form, the device uses Direct Current (DC) for its operation. A rotary component called an armature coil rests inside the motor's casing, surrounded by strong permanent magnets. When a current is applied to the armature through a rotary electric switch called a commutator, the magnetic field created by the armature interacts with the magnetic field of the stationary magnet to apply a torque on the armature, causing it to rotate.

3.1.3 MQ3 SENSOR

Fig 3.1.3 MQ 3 SENSOR

Tin Dioxide (SnO_2) is a thin layer that makes up the MQ-3 sensor. It is sorted out in a way that gives alcohol great affectability and benzene low affectivity. Because of its instantaneous driving circuit, it offers a dynamic response, superior quality, and a longer lifespan. It has a distinct interface style. Port pins 1, 2, and 3 on the sensor tend to the yield, GND, and VCC separately.

3.1.4 Battery

Fig 3.1.4: 9V BATTERY

The 9V battery is typically connected to the power input of the microcontroller and other electronic components, providing a steady stream of power to keep the system running. It is important to use a battery with sufficient voltage and capacity to ensure that the system operates reliably and consistently.

The use of a 9V battery in the alcohol engine lock system with MQ3 sensor has several advantages. Firstly, the 9V battery is compact and portable, making it easy to install and transport. Secondly, the 9V battery has a long shelf life, which means that it can be stored for extended periods without losing its charge. Thirdly, the 9V battery provides a stable and reliable source of power, ensuring that the system operates correctly and consistently.

3.1.5 Breadboard

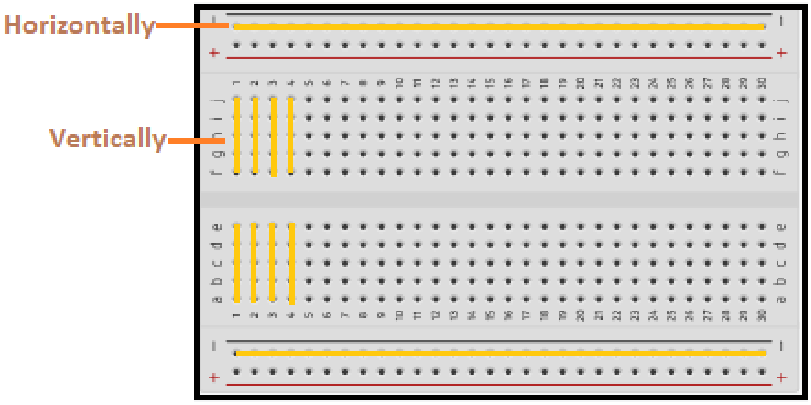

Fig 3.1.5. Breadboard

The breadboard is a white rectangular board with small embedded holes to insert electronic components. It is commonly used in electronics projects.

A breadboard is also categorized as a **Solderless board**. It means that the component does not require any soldering to fit into the board. Thus, we can say that breadboard can be reused. We can easily fit the components by plugging their end terminal into the board. Hence, a breadboard is often called a **plugboard**. The top and bottom holes of a row in a breadboard are connected horizontally, and the centre part is connected vertically

3.1.6 Motor driver

Fig3.1.6: Motor driver

L293D is a basic motor driver integrated chip (IC) that enables us to drive a DC motor in either direction and also control the speed of the motor. The L293D is a 16 pin IC, with 8 pins on each side, allowing us to control the motor. It means that we can use a single L293D to run up to two DC motors. L293D consist of two H-bridge circuit. H-bridge is the simplest circuit for changing polarity across the load connected to it.

There are 2 OUTPUT pins, 2 INPUT pins, and 1 ENABLE pin for driving each motor. It is designed to drive inductive loads such as solenoids, relays, DC motors, and bipolar stepper motors, as well as other high-current/high-voltage loads.

Technical Specification:

- Wide Supply-Voltage Range: 4.5 V to 36 V
- Output Current: 600 mA Per Channel
- Peak Output Current: 1.2 A Per Channel
- Supply Voltage to the IC: 4.5V to 7V
- Transition time: 300ns (at 5Vand 24V)
- Size: 44mm* 40mm*20mm (L*W*H)

Features:

- LM393 based design
- Easy to use with Microcontrollers or even with normal Digital/Analog IC
- Small, cheap and easily available
- Fixed bolt hole for easy installation
- Comparator output, clean signal, great waveform, and strong driving ability

Applications:

- Robotics Applications
- Mechatronics Applications

3.1.7 Wheels

Fig3.1.7. Wheel

Wheel Characteristics

1. Diameter: Ranges from 1-10 inches (2.5-25 cm), affecting speed and torque.

2. Width: Typically, 1-2 inches (2.5-5 cm), affecting stability and traction.

3. Hub Size: The size of the hub that attaches to the DC motor shaft.

4. Material: Affects durability, weight, and traction.

Common Wheel Sizes for DC Motors

1. Small: 1-2 inches (2.5-5 cm) diameter, often used for robotics and hobby projects.

2. Medium: 2-4 inches (5-10 cm) diameter, suitable for small to medium-sized projects.

3. Large: 4-6 inches (10-15 cm) diameter, often used for heavy-duty applications.

3.1.8 Jumper wires

Fig3.1.8. Jumper wires

Jumper wires are electrical wires used to connect two or more points in a circuit, allowing electricity to flow between them. They are commonly used in electronics, robotics, and other DIY projects.

Length: Jumper wires come in various lengths, from a few inches to several feet.

Gauge: Jumper wires are available in different gauges (thicknesses), with lower gauges indicating thicker wires.

Connectors: Jumper wires may have different types of connectors, such as male-female, male-male, or female-female.

3.1.9 Piezo Buzzer

Fig3.1.9. Piezo Buzzer

A piezo buzzer is a type of electronic device that's used to produce a tone, alarm or sound. It's lightweight with a simple construction as shown in fig 2.1.5, and it's typically a low-cost product. Yet at the same time, depending on the piezo ceramic buzzer specifications, it's also reliable and can be constructed in a wide range of sizes that work across varying frequencies to produce different sound outputs.

The piezo buzzer produces sound based on reverse of the piezoelectric effect. The generation of pressure variation or strain by the application of electric potential across a piezoelectric material is the underlying principle. These buzzers can be used alert a user of an event corresponding to a switching action, counter signal or sensor input. They are also used in alarm circuits.

The buzzer produces a same noisy sound irrespective of the voltage variation applied to it. It consists of piezo crystals between two conductors. When a potential is applied across these crystals, they push on one conductor and pull on the other. This, push and pull action, results in a sound wave. Most buzzers produce sound in the range of 2 to 4 kHz.

3.2 SOFTWARE REQUREMENT (ARDUINO IDE)

The ARDUINO Integrated Development Environment (IDE) is an application (for Windows, macOS, Linux) that is written in functions from C and C++. It contains a text editor for writing code, a message area, a text console, a toolbar with buttons for common functions and a series of menus. It connects to the Arduino and Genuine hardware to upload programs and communicate with them.**29** Programs written using Arduino Software (IDE) are called sketches. These sketches are written in the text editor and are saved with the file extension. ino the editor has features for cutting/passing and for searching/replacing text. The message area gives feedback while saving and exporting and also display errors. The console displays a text output by the Arduino Software (IDE), including complete error messages and other information. The bottom righthand corner of the window displays the configured board and serial port. The toolbar buttons allow you to verify and upload programs, create, open, and save sketches, and open the serial monitor. Versions of the Arduino Software (IDE) prior to 1.0 saved sketches with the extension.pde. It is possible to open these files with version 1.0, you will be promoted to save the sketches with the .ino extension on save. Here below some of the function in the IDE are discussed.

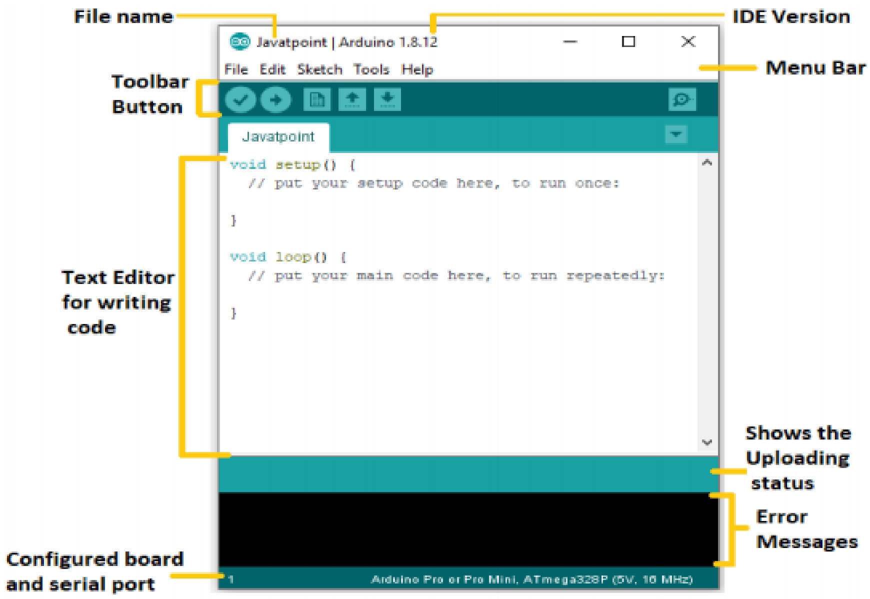

Fig3.2.1. Arduino Uno

3.2.1 Sketchbook

The Arduino Software (IDE) uses the concept of a sketchbook: a standard place to store your programs (or sketches). The sketches in your sketchbook can be opened from the File > Sketchbook menu or from the Open button on the toolbar as shown in above Fig 3.2.1. The first time you run the Arduino software, it will automatically create a directory for your sketchbook. You can view or change the location of the sketchbook location from with the Preferences dialog. Beginning with version 1.0, files are saved with a .ino file extension. Previous versions use the .pde extension. You may still open .pde named files in version 1.0 and later, the software will automatically rename the extension to .ino.

3.2.2 Serial Monitor

This displays serial sent from the Arduino or Genuino board over USB or serial connector. To send data to the board, enter text and click on the "send" button or press enter. Choose the baud rate from the drop-down menu that matches the rate passed to Serial. begin in your sketch. Note that on Windows, Mac or Linux the board will reset (it will rerun your sketch) when you connect with the serial monitor. Please note that the Serial Monitor does not process control characters; if your sketch needs a complete management of the serial communication with control characters, you can use an external terminal program and connect it to the COM port assigned to your Arduino board.

CHAPTER 4

4.DESIGN AND IMPLEMENTATION

4.1 System Block Diagram

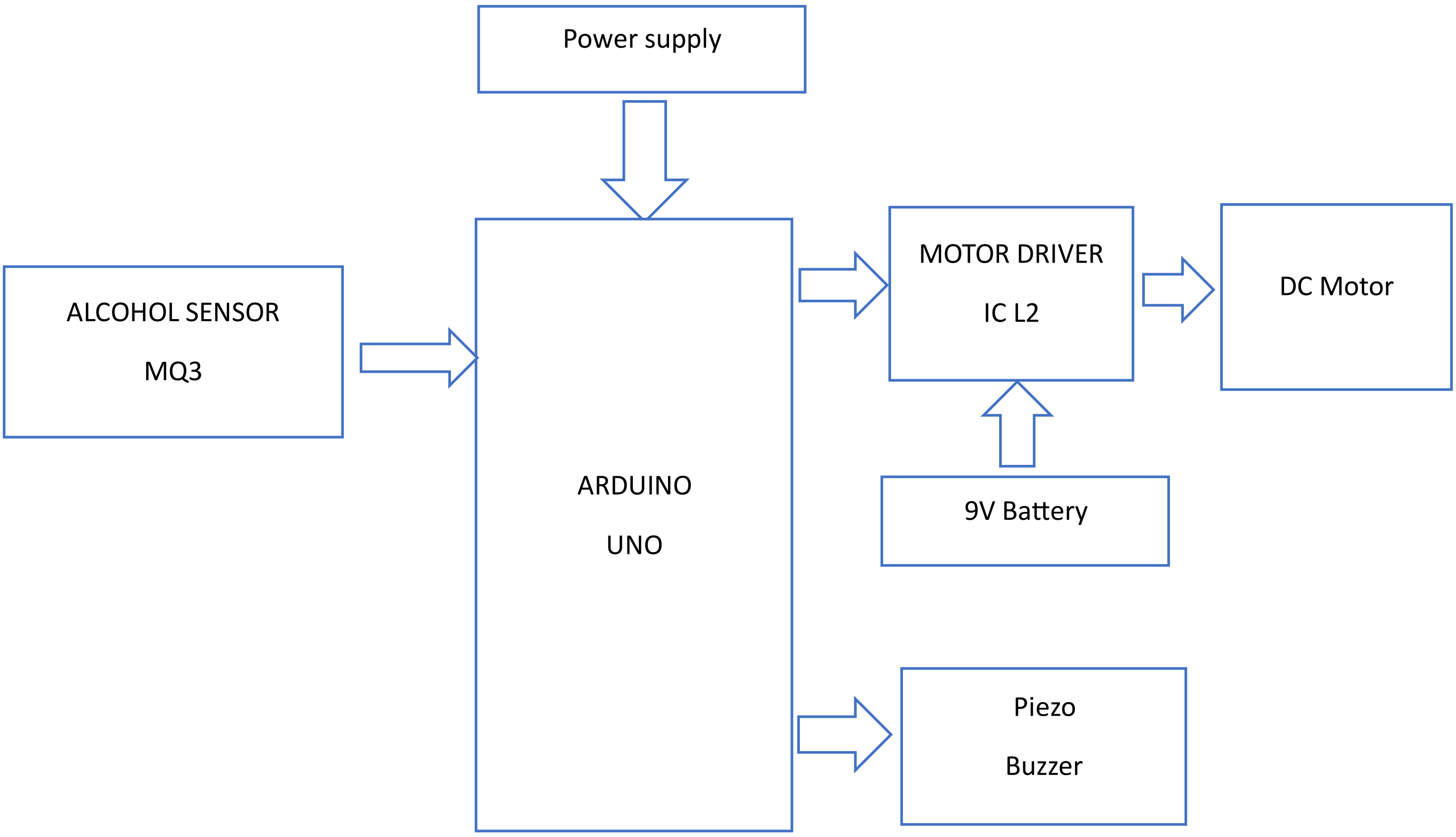

FIG 4.1. Block diagram

System block diagram is comprised of: Alcohol detection engine lock using (Arduino UNO):related to alcohol detection and alert the driver with the components used in the proposed operation are MQ3 Alcohol sensor, power supply, Arduino (UNO) ,Piezo Buzzer, LED, Motor Driver IC L293d,DC motor, and 9v battery. The main component is Arduino Uno which is an ATmega328 based microcontroller (MC) that performs all functions related to controlling the embedded system circuit.

4.2 Circuit diagram

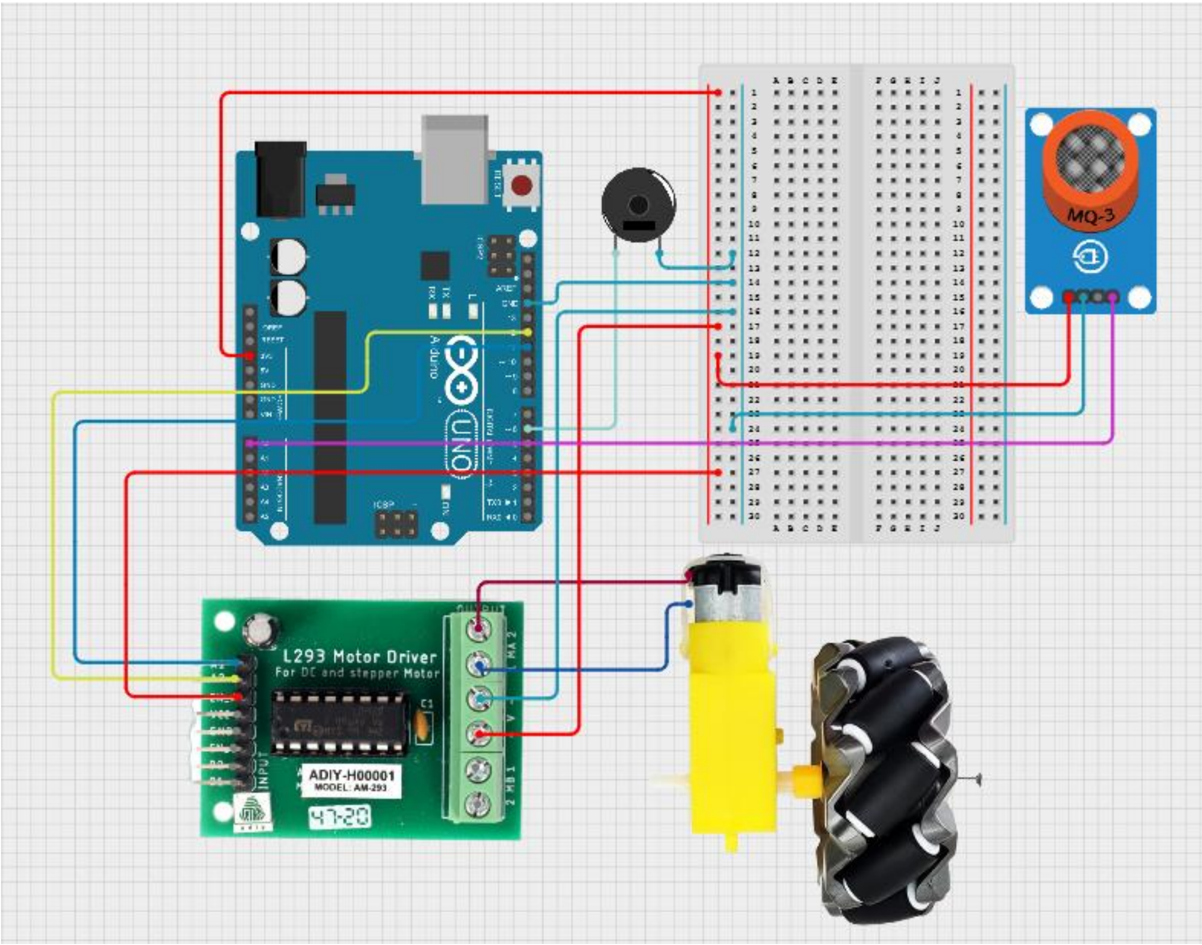

Fig4.2. Circuit Diagram

4.3 CODE

```
int alcoholPin = A0;  // MQ-3 Alcohol sensor connected to A0
int motorPin1 = 11;    // Motor control pin 1 (IN1)
int motorPin2 = 12;    // Motor control pin 2 (IN2)
int buzzerPin = 6;     // Buzzer connected to Pin 6
int threshold = 500;   // Threshold alcohol level for triggering motor lock
int alcoholLevel = 0;  // Variable to store alcohol sensor reading

void setup() {
  pinMode(motorPin1, OUTPUT);   // Set motor control pins as output
  pinMode(motorPin2, OUTPUT);
  pinMode(buzzerPin, OUTPUT);    // Set buzzer pin as output

  Serial.begin(9600);           // Start Serial Monitor for debugging
  Serial.println("Alcohol Detection System Initialized...");
  Serial.println("Threshold Level: ");
  Serial.println(threshold);    // Print the threshold level for reference
}

void loop() {
  // Read alcohol level from MQ-3 sensor
  alcoholLevel = analogRead(alcoholPin);
  Serial.print("Alcohol Sensor Reading: ");
  Serial.println(alcoholLevel);  // Display raw alcohol level reading on Serial Monitor



  // Compare the alcohol level with the threshold
```

```
  if (alcoholLevel > threshold) {
    // Alcohol detected above threshold: stop motor and activate buzzer
    Serial.println("ALCOHOL DETECTED! Exceeding threshold level.");
    Serial.println("Motor stopped, Buzzer ON.");

    digitalWrite(motorPin1, LOW);  // Stop motor
    digitalWrite(motorPin2, LOW);
    digitalWrite(buzzerPin, HIGH); // Activate buzzer

  }
else
 {

    // No alcohol detected: run motor and keep buzzer off
    Serial.println("No alcohol detected. Level is below the threshold.");
    Serial.println("Motor running...");

    digitalWrite(motorPin1, HIGH);  // Run motor forward
    digitalWrite(motorPin2, LOW);
    digitalWrite(buzzerPin, LOW);   // Turn off buzzer

}



  // Print the current system state to the Serial Monitor

  Serial.print("Motor Status: ");
  if (alcoholLevel > threshold) {
    Serial.println("Stopped");
  }
```

```
 else
 {
   Serial.println("Running");
  }

  Serial.print("Buzzer Status: ");
  if (digitalRead(buzzerPin) == HIGH) {
    Serial.println("ON");
  } else {
    Serial.println("OFF");
  }

  Serial.println();  // Add a blank line between readings for better readability
  delay(2000);  // Delay for 1 second before the next sensor reading
}
```

CHAPTER 5

5 Test & Result

5.1 Result

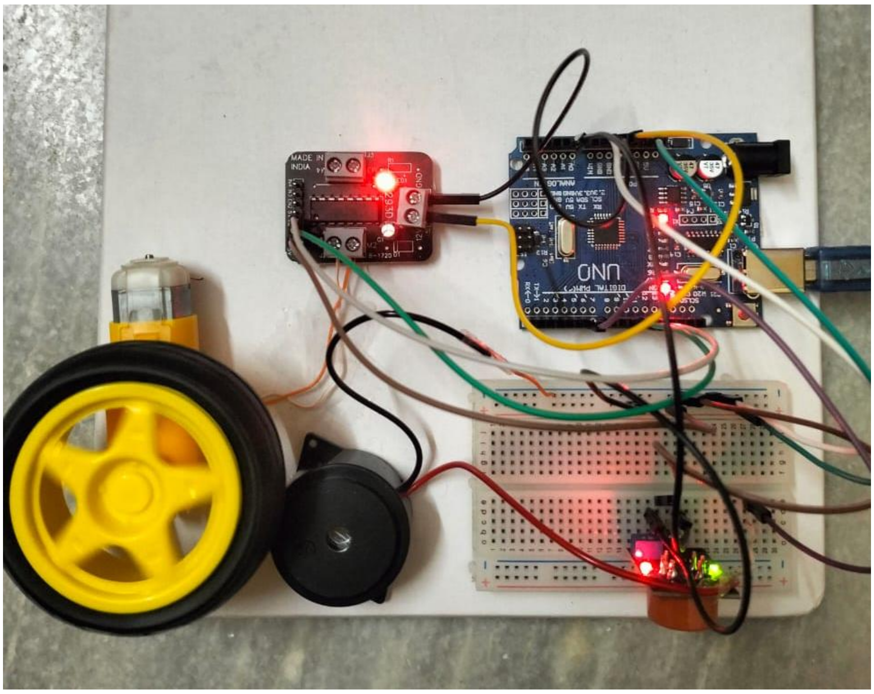

Fig 5.1 Result

When no alcohol is detected

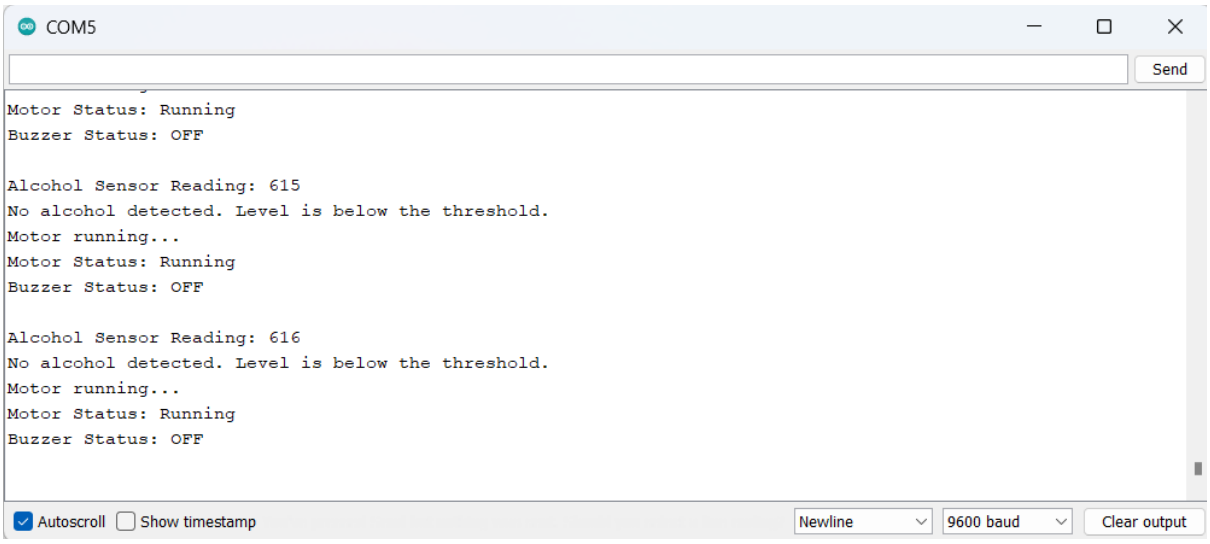

Fig 5.1.1 Alcohol not detected (serial monitor)

As the alcohol sensor reading is below the threshold value i.e., 700, the motor is running and the buzzer is off.

When alcohol is detected

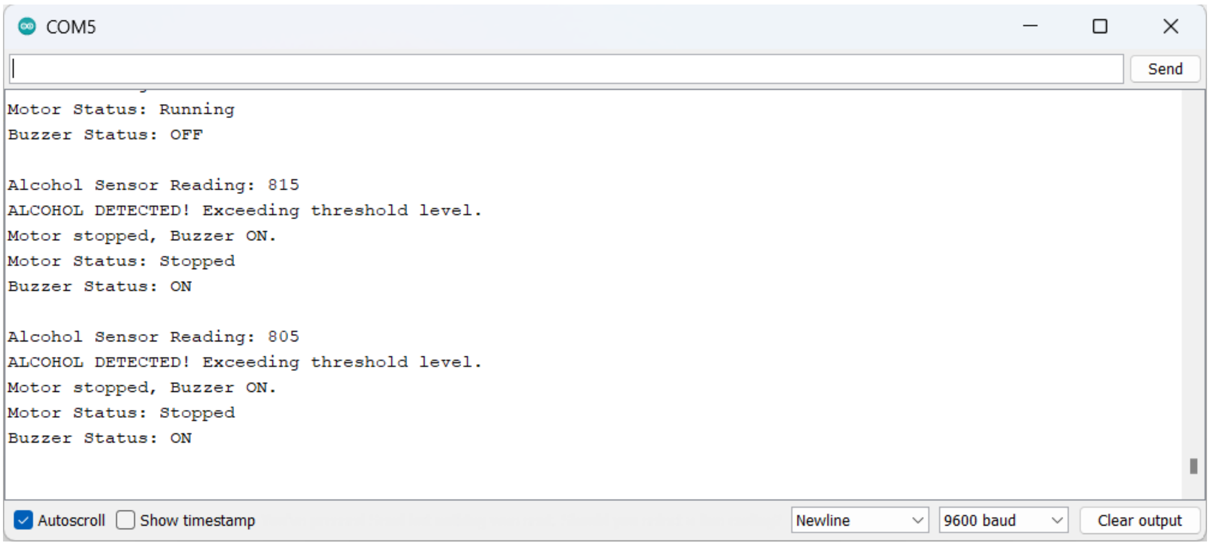

Fig 5.1.2. Alcohol detected (serial monitor)

As the alcohol sensor reading is above the threshold value i.e., 700, the motor is turned off and the buzzer is turned on.

5.2. FUTURE DEVELOPMENTS

The development of alcohol detectors and engine locking systems is an ongoing process, and there are several potential future developments that could enhance their effectiveness and functionality. Here are some of the possibilities: Integration with biometric sensors: Future alcohol detectors and engine locking systems may incorporate biometric sensors to detect specific physical characteristics that indicate impairment, such as changes in pupil size, body temperature, or heart rate. Real-time monitoring and reporting: Advanced alcohol detectors and engine locking systems may be capable of transmitting data in real-time to law enforcement agencies or other relevant parties, enabling immediate action to be taken if a driver is found to be impaired. Improved accuracy: New sensor technologies may be developed that can detect alcohol at lower levels or more accurately differentiate between alcohol and other substances, such as mouthwash or hand sanitizer.

Artificial intelligence (AI) integration: Future systems may incorporate AI algorithms to analyze data from multiple sources, including biometric sensors, GPS, and other vehicle sensors, to better detect impairment and prevent false positives. Wireless connectivity: Alcohol detectors and engine locking systems may be equipped with wireless connectivity, allowing them to communicate with other vehicle systems or with external devices, such as smartphones or wearable devices. User customization: Advanced systems may allow users to customize their settings based on their individual needs or preferences, such as setting a different threshold for impairment or disabling the engine lock feature under certain circumstances. Overall, the future of alcohol detectors and engine locking systems is likely to involve a combination of new sensor technologies, advanced data analytics, and improved user interfaces, all aimed at improving their accuracy, effectiveness, and ease of use.

CHAPTER 6

6.1 ADVANTAGES

The alcohol detection with an engine locking system can be implemented in any two or four wheelers

• The Government can keep track of drunken driving cases

• It can provide quick and accurate results • Change of loss of life and property is minimized

6.2 APPLICATIONS

Automatic engine locking systems through alcohol detection projects can be used in various vehicles for detecting whether the driver has consumed alcohol or not.

• This project can be used in various companies, organizations, and mines to detect the alcohol consumption of employees

6.3 CONCLUSION

In conclusion, the alcohol detector and engine locking system is an important safety feature that can potentially save lives and prevent accidents. It accurately measures a driver's BAC level and prevents them from operating their vehicle if their level is above the legal limit. However, the system also has limitations, such as the possibility of drivers attempting to bypass it and the cost of installation. Overall, the alcohol detector and engine locking system is a valuable tool in preventing drunk driving, but it is not a substitute for responsible driving behaviour and public education campaigns about the dangers of drunk driving.

6.4 REFERENCES

https://www.irjet.net/archives/V10/ICRTET23/IRJET-V10I450.pdf

https://nevonprojects.com/arduino-based-alcohol-sense-engine-lock-gps/

https://www.youtube.com/watch?v=KqIvraVYbZ4

CHAPTER 7

PPT Slides

ARDUINO BASED ALCOHOL SENSE ENGINE LOCK

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UNDER THE GUIDENCE OF

S.Sreehari Ast. Prof.

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- Introduction
- Advantages and Disadvantages
- Working Operation of the circuit
- Specifications
- Circuit Diagram
- Tools for Implementation
- Implementation plan during semester

OBJECTIVES

The main objective of an Arduino based alcohol sense engine lock is to enhance road safety by preventing drunk driving incidents by disabling the engine when alcohol is detected.

INTRODUCTION

- Nowadays most road accidents are caused due to drunk driving. The drunk drivers are not in stable condition to drive the vehicle safely.
- This may lead to accidents. The Arduino-based alcohol sense engine lock has far-reaching potential across various industries, prioritizing safety, security, and responsible driving practices.
- This is an innovative solution to prevent drunk driving. We Integrate an alcohol sensor with an Arduino board.

Advantages and Disadvantages

Advantages

- Actively prevents drunk driving
- Reliable and tamper-resistant design
- Enhances driver and public safety
- Can be integrated into new or existing vehicles

Disadvantages

- Potential inconvenience for sober drivers
- Requires regular calibration and maintenance
- Potential for false positives or malfunctions
- Increased cost for vehicle installation

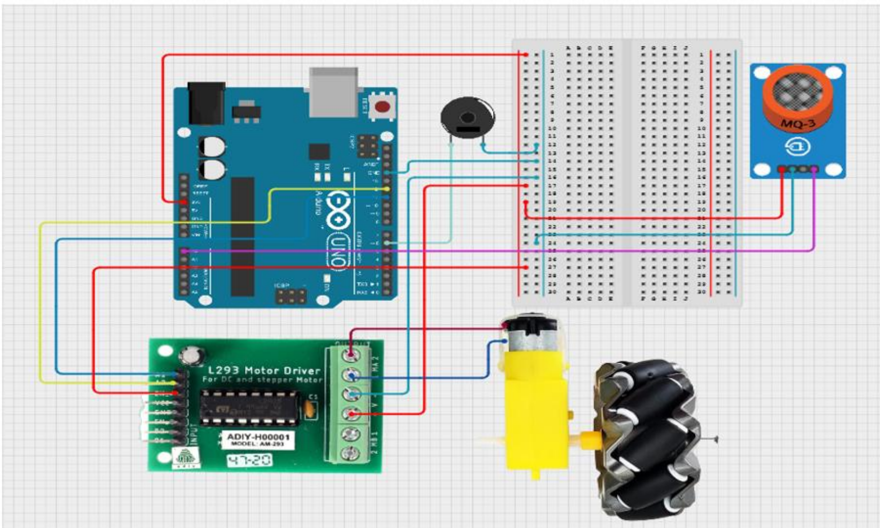
Working operation of the circuit

- First power supply has to be given through the lithium ion battery.
- The system detects the breath alcohol levels of the person riding the vehicle.
- When a limited threshold value is reached the engine will be locked.

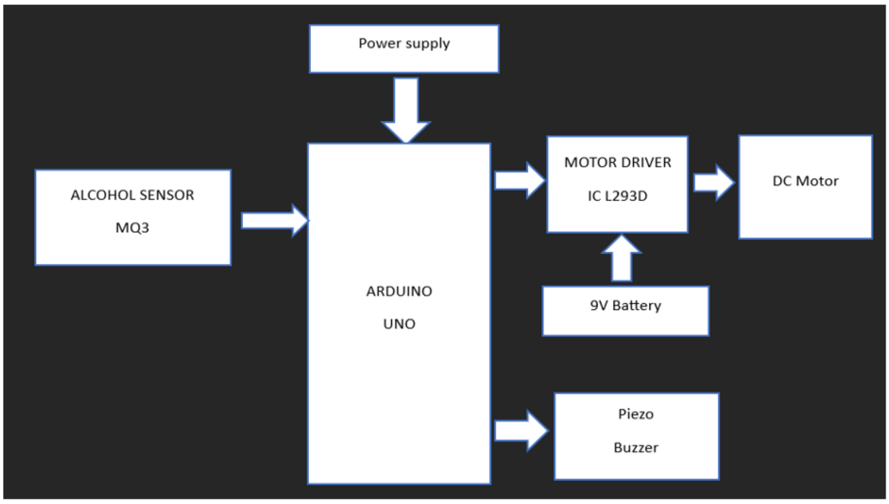
Specifications

- Arduino UNO board
- Alcohol Sensor MQ-3
- Battery
- Buzzer
- DC motor
- Ply board
- Jumper wires
- Breadboard
- Wheels

Circuit diagram



Block Diagram



Tools for implementation

1. Arduino IDE (Integrated Development Environment)
2. Programming languages: C/C++ (for Arduino)

Implementation plan during semester

- 18-9-2024 to 28-9-2024 → Selection of mini project topic
- 29-9-2024 to 12-10-2024 → Literature survey of specifications
- 13-10-2024 to 25-10-2024 → Mini project abstract
- 26-10 -2024 to 10-11-2024→ Components purchasing
- 11-11-2024 to 30-12-2024→ Project execution
- 31-12-2024 to 03-01-2025→ Results verification
- 01-01-2024 to 20-01-2025→ Report preparation
- January 2025 → Report submission

Conclusion

- This project can be used to encourage drivers to make informed decisions and avoid operating a vehicle under the influence.
- To reduce liability and to Protect vehicle owners and businesses from legal and financial consequences of drunk driving.
- To reduce drunk driving crashes that often result in serious injuries, fatalities, and immense emotional and financial costs for victims and their families.